certain states and foreign jurisdictions to fully utilize available tax credits and other attributes. The deferred tax assets are offset by a valuation allowance to the extent it is more likely than not that they are not expected to be realized.

We provide U.S. income taxes on the earnings of foreign subsidiaries unless the subsidiaries' earnings are considered permanently reinvested outside the United States or are exempted from taxation as a result of the new territorial tax system. To the extent that the foreign earnings previously treated as permanently reinvested are repatriated, the related U.S. tax liability may be reduced by any foreign income taxes paid on these earnings. As of November 30, 2018, the cumulative amount of earnings upon which U.S. income taxes have not been provided is approximately $275 million. The unrecognized deferred tax liability for these earnings is approximately $57.8 million.

As of November 30, 2018, we have net operating loss carryforwards of approximately $881.1 million for federal and $349.7 million for state. We also have federal, state and foreign tax credit carryforwards of approximately $8.8 million, $189.9 million and $14.9 million, respectively. The net operating loss carryforward assets and tax credits will expire in various years from fiscal 2019 through 2036. The state tax credit carryforwards and a portion of the federal net operating loss carryforwards can be carried forward indefinitely. The net operating loss carryforward assets and certain credits are reduced by the valuation allowance and are subject to an annual limitation under Internal Revenue Code Section 382, the carrying amount of which are expected to be fully realized.

As of November 30, 2018, a valuation allowance of $174.5 million has been established for certain deferred tax assets related to certain state and foreign assets. For fiscal 2018, the total change in the valuation allowance was $80.9 million.

Accounting for Uncertainty in Income Taxes

During fiscal 2018 and 2017, our aggregate changes in our total gross amount of unrecognized tax benefits are summarized as follows (in thousands):

	2018	2017
Beginning balance	$ 172,945	$ 178,413
Gross increases in unrecognized tax benefits – prior year tax positions	16,191	3,680
Gross decreases in unrecognized tax benefits – prior year tax positions	(4,000)	(30,166)
Gross increases in unrecognized tax benefits – current year tax positions	60,721	24,927
Settlements with taxing authorities	—	(3,876)
Lapse of statute of limitations	(45,922)	(8,819)
Foreign exchange gains and losses	(3,783)	8,786
Ending balance	$ 196,152	$ 172,945

The combined amount of accrued interest and penalties related to tax positions taken on our tax returns were approximately $24.6 million and $23.6 million for fiscal 2018 and 2017, respectively. These amounts were included in long-term income taxes payable in their respective years.

We file income tax returns in the United States on a federal basis and in many U.S. state and foreign jurisdictions. We are subject to the continual examination of our income tax returns by the IRS and other domestic and foreign tax authorities. Our major tax jurisdictions are Ireland, California and the United States. For Ireland, California and the United States, the earliest fiscal years open for examination are 2008, 2014 and 2015, respectively. We regularly assess the likelihood of outcomes resulting from these examinations to determine the adequacy of our provision for income taxes and have reserved for potential adjustments that may result from these examinations. We believe such estimates to be reasonable; however, there can be no assurance that the final determination of any of these examinations will not have an adverse effect on our operating results and financial position.

The timing of the resolution of income tax examinations is highly uncertain as are the amounts and timing of tax payments that are part of any audit settlement process. These events could cause large fluctuations in the balance of short-term and long-term assets, liabilities and income taxes payable. We believe that within the next 12 months, it is reasonably possible that either certain audits will conclude or statutes of limitations on certain income tax examination periods will expire, or both. Given the uncertainties described above, we can only determine a range of estimated potential effect in underlying unrecognized tax benefits ranging from $0 to approximately $45 million.